

GRAVESEND-BROADNESS, United Kingdom

WMO: 037840

Lat: 51.4645N

Lon: 0.3113E

Elev: 10

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 96-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.5	29.8	22.5	16.9	31.7	24.9	19.0	33.2	29.8	50.6	27.1	50.0	6.1	160	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	82.3	67.3	78.5	65.5	75.3	63.8	69.1	79.1	67.0	75.5	65.3	72.8	8.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	93.9	72.8	63.7	88.3	70.9	62.2	83.7	69.3	33.1	79.3	31.5	75.4	30.2	72.9	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.8	22.2	19.6	DB	22.3	89.4	3.0	3.7	20.2	92.1	18.4	94.2	16.7	96.3	14.5	99.0	
			WB	21.7	72.4	3.3	3.0	19.3	74.6	17.3	76.4	15.4	78.1	13.0	80.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	41.9	42.3	45.4	49.8	55.5	61.2	65.2	65.0	60.2	54.2	47.3	42.6	(6)
	DBStd	9.95	6.02	5.73	5.40	5.40	5.07	4.62	4.40	4.34	4.33	4.96	5.57	6.28	(7)
	HDD50	1087	256	221	158	68	11	0	0	0	0	19	114	240	(8)
	HDD65	4667	716	636	606	457	296	132	53	53	155	337	531	695	(9)
	CDD50	2036	5	5	17	61	181	336	470	464	305	148	33	11	(10)
	CDD65	141	0	0	0	0	2	17	59	52	10	1	0	0	(11)
	CDH74	990	0	0	0	6	23	155	392	341	65	8	0	0	(12)
	CDH80	211	0	0	0	0	2	34	74	90	9	2	0	0	(13)
(14) Wind	WSAvg	9.7	10.9	11.3	10.5	9.6	9.9	9.0	8.6	8.5	8.7	9.4	9.7	10.4	(14)
(15) Precipitation	PrecAvg	25.7	2.6	2.0	1.5	1.9	1.9	1.8	1.8	2.1	2.0	2.7	3.0	2.5	(15)
	PrecMax	36.7	6.3	4.6	4.0	4.9	4.7	4.6	3.8	3.8	5.0	7.2	6.7	5.2	(16)
	PrecMin	18.3	0.4	0.3	0.2	0.0	0.4	0.4	0.4	0.1	0.4	0.4	1.1	0.4	(17)
	PrecStd	4.6	1.4	1.2	0.9	1.3	1.0	1.3	0.8	1.0	1.1	1.5	1.5	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	57.6	63.8	73.3	76.8	84.6	85.9	87.1	80.7	71.2	62.0	57.9	(19)
		MCWB	53.0	52.1	54.3	60.3	64.7	67.6	68.6	69.3	65.7	63.3	58.8	54.7	(20)
	2%	DB	54.7	54.6	59.2	66.4	72.5	78.2	81.9	81.9	74.9	65.9	59.4	55.9	(21)
		MCWB	51.7	50.4	51.7	56.0	61.9	65.1	67.1	67.4	64.2	59.4	55.9	53.3	(22)
	5%	DB	52.7	52.6	56.4	62.5	68.9	74.1	78.4	77.3	71.1	63.7	57.5	54.2	(23)
		MCWB	49.9	48.9	50.3	53.5	59.5	62.8	65.3	65.3	62.3	58.5	54.1	51.6	(24)
	10%	DB	50.8	50.8	54.1	59.4	65.5	70.8	75.1	73.6	68.4	61.7	55.8	52.3	(25)
		MCWB	48.2	47.4	49.2	51.9	57.1	61.1	63.8	63.6	60.7	57.3	52.8	49.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	53.3	55.6	60.6	66.6	69.9	72.5	71.2	68.3	63.5	59.3	55.6	(27)
		MCDB	55.7	56.4	61.4	71.4	74.4	81.0	81.1	84.2	76.2	70.2	61.2	57.0	(28)
	2%	WB	52.3	51.2	53.1	57.0	63.1	66.5	69.0	68.8	65.7	61.2	56.6	53.7	(29)
		MCDB	54.4	53.7	57.6	64.1	70.3	74.8	78.6	79.0	72.4	64.4	58.8	55.7	(30)
	5%	WB	50.2	49.5	51.5	54.7	60.5	64.1	66.6	66.8	63.7	59.8	54.8	51.8	(31)
		MCDB	52.3	52.0	55.4	60.6	67.2	72.0	75.5	74.7	69.4	62.7	57.0	53.9	(32)
	10%	WB	48.5	47.8	49.7	52.8	58.3	62.1	64.8	64.9	62.0	58.2	52.9	49.7	(33)
		MCDB	50.7	50.3	53.4	58.1	64.2	69.5	72.8	71.5	66.8	61.0	55.3	52.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	10.1	12.5	15.6	16.1	16.6	16.9	16.2	15.4	12.5	10.0	8.6	(35)
		MCDBR	9.8	11.5	16.9	21.3	22.2	22.7	23.0	22.5	19.7	14.4	10.2	8.8	(36)
	5% WB	MCWBR	9.1	8.6	10.8	12.1	12.9	11.8	10.8	10.2	10.6	9.4	8.4	8.1	(37)
		MCWBR	9.6	10.5	15.3	19.5	20.1	20.5	20.7	19.9	17.8	12.7	9.7	8.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.334	0.349	0.387	0.407	0.413	0.414	0.411	0.401	0.382	0.365	0.338	0.324	(40)
		taud	2.369	2.371	2.260	2.228	2.254	2.277	2.296	2.342	2.390	2.418	2.414	2.360	(41)
	Ebn at Noon		213	244	254	263	267	267	266	262	254	234	210	195	(42)
		Edh at Noon	20	26	35	40	41	40	39	36	30	24	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		251	462	817	1303	1554	1726	1638	1364	1024	587	301	194	(44)
	RadStd		23	68	104	128	109	142	147	106	82	55	35	25	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page